DIGITAL LOGIC FUNDAMENTALS

Course Code	:30801220
Credit Hours	:3
Prerequisite	: 30801101

Course description {Numbering systems, Boolean algebra, logic algebra, basic logic gates, minimization of logic functions, combinational logic: adders, subtractors, encoders and decoders, multiplexers and demultiplexers, sequential logic: flip-flops, counters, registers and clocked sequential circuits.
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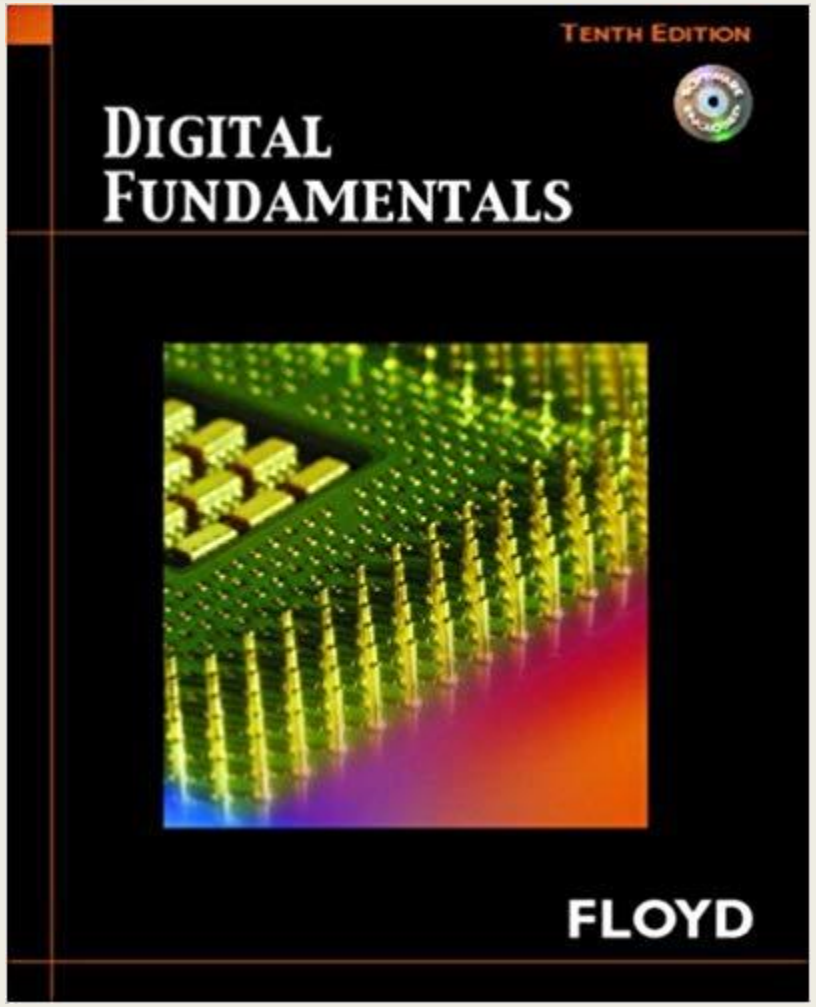
Course Title; Digital Logic Fundamentals

Credit Hour(3)

[Pre-req. 30801101**]**

Textbook: Textbook Title



COURSE OBJECTIVES

The course consists of a set of experiments designed to allow students to build, and verify digital circuits and systems. This set of experiments cover relevant topics designed to reinforce the theoretical concepts taught in the classroom with practical experience in the lab. By the end of the course, students are expected to have a good understanding of digital logic design.

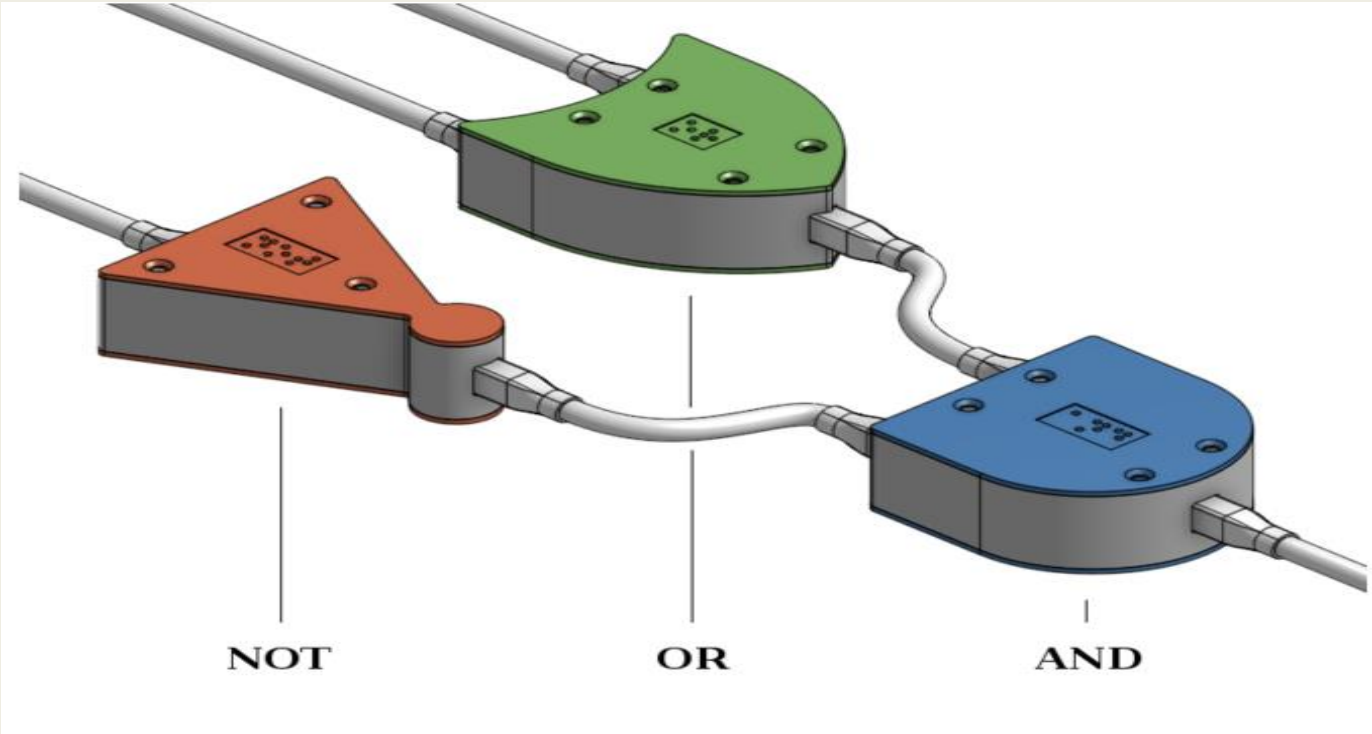
COURSE SYLLABUS

Week	Course Topic
Week 1	Chapter 1, Digital logic gates
Week 2	
Week 3	
Week 4	
Week 5	
Week 6	
Week 7	
Week 8	Midterm Exam
Week 9	
Week 10	
Week 11	
Week 12	
Week 13	
Week 14	
Week 15	
Week 16	Final Exam

Chapter 1

Digital Logic Gates

LOGIC GATES



Digital Logic

- Digital electronics uses circuits that have two states, which are represented by two different voltage levels called HIGH and LOW.
- The voltages represent numbers in the binary system.
- In binary, a single number is called a *bit* (for *b*inary dig*it*).
- A bit can have the value of either a 0 or a 1, depending on if the voltage is HIGH or LOW.

Digital Logic

- Boolean algebra is a mathematical system for the manipulation of variables that can have one of two values.
  - *In formal logic, these values are “true” and “false.”*
  - *In digital systems, these values are “on” and “off,” 1 and 0, or “high” and “low.”*
- Boolean expressions are created by performing operations on Boolean variables.
  - *Common Boolean operators include AND, OR, and NOT.*

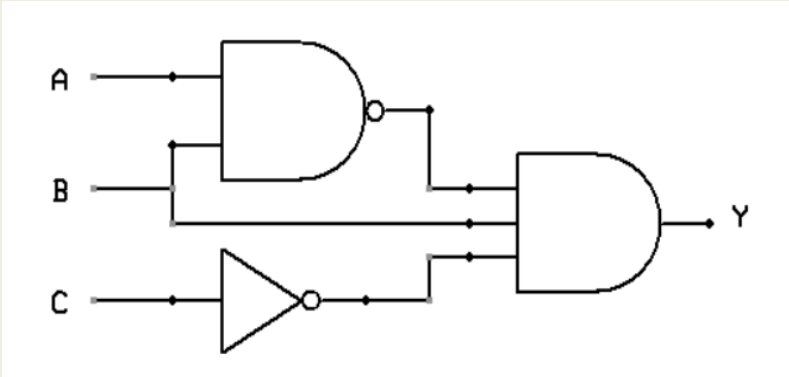
Five ways (representations) to represent the logic:

1. Boolean expression in equation form

   (POS) & (SOP) forms

$$Y = \left(\overline{AB}\right)B\overline{C}$$

2. Logic circuit diagram

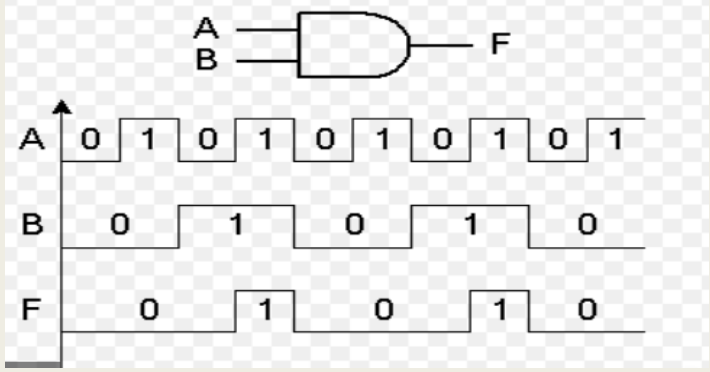


Logic representation

3. Truth Tables

Inputs		Output
A	B	L
0	0	0
0	1	1
1	0	1
1	1	0

4. Timing Diagram



Logic representation (Continued)

5. Karnaugh Maps

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	0	0	1	1
10	1	0	1	1

- We can convert from form to another in a systematic way

Logic gates

- Digital electronics relies on the actions of seven types of logic gates:

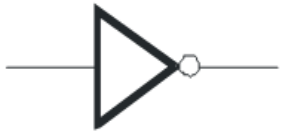
1. *AND,*
2. *OR,*
3. *NAND (Not AND),*
4. *NOR (Not OR),*
5. *XOR (Exclusive OR)*
6. *XNOR (Exclusive NOR)*
7. *NOT*

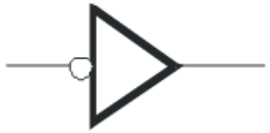
Basic Logic Functions

AND True only if ***all*** input conditions are true.

OR True only if ***one or more*** input conditions are true.

NOT Indicates the ***opposite*** condition.





Logic gates principle

- **And**, **or**, and **not** elements can be combined to form various logic functions and arithmetic functions

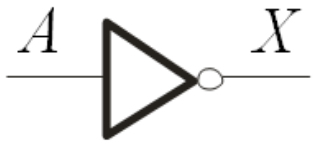
- Gate: A logic circuit that performs a basic logic operations such as AND or OR , etc

- Each type of gate has one or more (most often two) inputs and one output.

- The principle of operation is that the circuit operates on just two voltage levels, called logic 0 and logic 1.

- When either of these voltage levels is applied to the inputs, the output of the gate responds by assuming a 1 or a 0 level, depending on the logic of the gate.

The Inverter



The inverter performs the Boolean **NOT** operation. When the input is LOW, the output is HIGH; when the input is HIGH, the output is LOW.

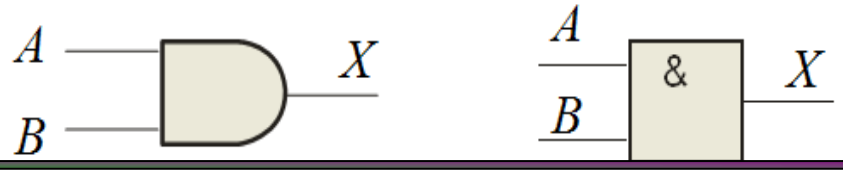
Input	Output
A	X
LOW (0)	HIGH (1)
HIGH (1)	LOW(0)

The **NOT** operation (complement) is shown with an overbar. Thus, the Boolean expression for an inverter is $X = \overline{A}$.

Complement
Overbar symbol

Truth table: A table showing the inputs and corresponding output(s) of a logic circuit.

The AND Gate



The **AND gate** produces a HIGH output when all inputs are HIGH; otherwise, the output is LOW. For a 2-input gate, the truth table is

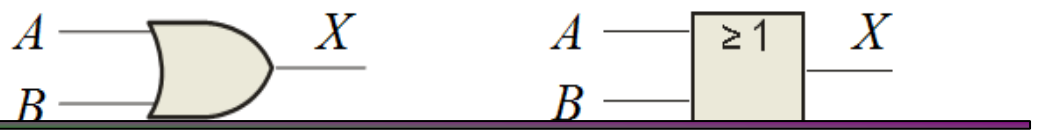
Inputs		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

All inputs high = 1

The **AND** operation is usually shown with a dot between the variables but it may be implied (no dot). Thus, the AND operation is written as $X = A \cdot B$ or $X = AB$.

Dot or no dot between variables

The OR Gate



The **OR gate** produces a HIGH output if any input is HIGH; if all inputs are LOW, the output is LOW. For a 2-input gate, the truth table is

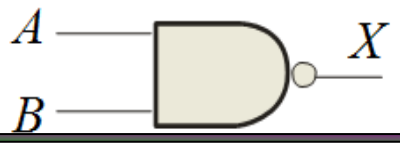
Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

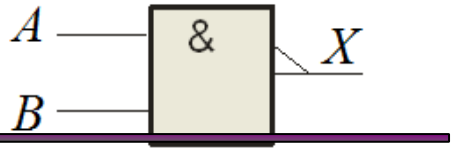
Any input is high = 1

The **OR** operation is shown with a plus sign (+) between the variables. Thus, the OR operation is written as $X = A + B$.

Plus sign

The NAND Gate





The **NAND gate** produces a LOW output when all inputs are HIGH; otherwise, the output is HIGH. For a 2-input gate, the truth table is

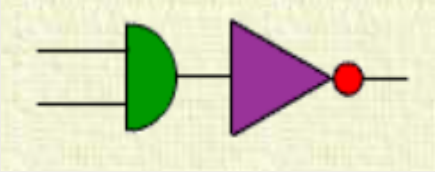
Inputs		Output
A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

All inputs high = 0

The **NAND** operation is shown with a dot between the variables and an overbar covering them. Thus, the NAND operation is written as $X = \overline{A \cdot B}$ (Alternatively, $X = \overline{AB}$.)

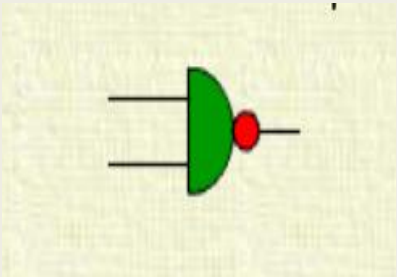
- NAND Gate

And symbol then the inverter symbol

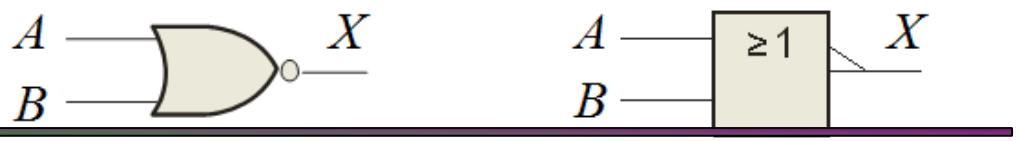


- OR

And symbol followed by a bubble



The NOR Gate



The **NOR gate** produces a LOW output if any input is HIGH; if all inputs are HIGH, the output is LOW. For a 2-input gate, the truth table is

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

The **NOR** operation is shown with a plus sign (+) between the variables and an overbar covering them. Thus, the NOR operation is written as $X = \overline{A + B}$.

The XOR Gate

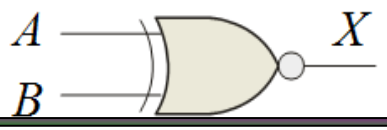
The **XOR gate** produces a HIGH output only when both inputs are at opposite logic levels. The truth table is

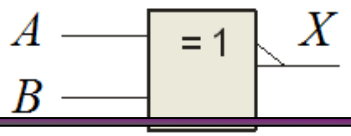
Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	0

Opposite logic level = 1

The **XOR** operation is written as $X = \overline{A}B + A\overline{B}$. Alternatively, it can be written with a circled plus sign between the variables as $X = A \oplus B$.

The XNOR Gate





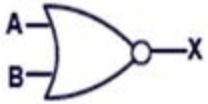
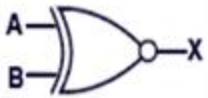
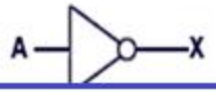
The **XNOR gate** produces a HIGH output only when both inputs are at the same logic level. The truth table is

Inputs		Output
A	B	X
0	0	1
0	1	0
1	0	0
1	1	1

Same logic level = 1

The **XNOR** operation shown as $X = \overline{A}\,\overline{B} + AB$. Alternatively, the XNOR operation can be shown with a circled dot between the variables. Thus, it can be shown as $X = A \odot B$.

Summary

ANSI Symbol	Description	Boolean
	The AND gate output is at logic 1 when, and only when all its inputs are at logic 1, otherwise the output is at logic 0.	$X = A \cdot B$
	The OR gate output is at logic 1 when one or more of its inputs are at logic 1. If all the inputs are at logic 0, the output is at logic 0.	$X = A + B$
	The NAND Gate output is at logic 0 when, and only when all its inputs are at logic 1, otherwise the output is at logic 1.	$X = \overline{A \cdot B}$
	The NOR gate output is at logic 0 when one or more of its inputs are at logic 1. If all the inputs are at logic 0, the output is at logic 1.	$X = \overline{A + B}$
	The XOR gate output is at logic 1 when one and ONLY ONE of its inputs is at logic 1. Otherwise the output is logic 0.	$X = A \oplus B$
	The XNOR gate output is at logic 0 when one and ONLY ONE of its inputs is at logic 1. Otherwise the output is logic 1. (It is similar to the XOR gate, but its output is inverted).	$X = \overline{A \oplus B}$
	The NOT gate output is at logic 0 when its only input is at logic 1, and at logic 1 when its only input is at logic 0. For this reason it is often called an INVERTER.	$X = \overline{A}$